Universiteit Antwerpen
Faculteit Toegepaste Ingenieurswetenschappen

5 – Infrastructuur & Beveiliging

Lab 3: Spanning tree

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1 Preparation

Use the material below to prepare for the lab and for the short multiple-choice test at the start of the lab session. The goal is to **understand the concepts**, not to memorize. You will **not** be asked to reproduce commands, configuration syntax, or specific numbers such as timer values. Instead, you should be able to explain what happens in a switched network with redundant links, and why.

Start with the NetworkAcademy.io lessons on the Spanning Tree Protocol. They cover why loops are a problem in Ethernet networks, how STP builds a loop-free topology, and how it reacts to changes. You do not have to study the whole course; study the following lessons:

- Why do we need Spanning-Tree? [1]
- The Spanning-Tree Algorithm [2]
- Spanning-Tree Port Roles[1] [3]
- STP Port States and Timers [4]
- STP Topology Changes (TCN) [5]

The examples use Cisco switches, so the notation on our OpenWrt switches looks slightly different, but the mechanism is the same. Finally, read Types of Spanning Tree Protocol to see how classic STP relates to its successors [6].

If you prefer a more visual explanation, watch the Spanning Tree Protocol video; the first 8.5 minutes provide a visual walkthrough of the STP algorithm [7], but the video is not a substitute for the required reading

2 Lab Goals

In this lab, you will build redundant Layer 2 networks and use the Spanning Tree Protocol (STP) to prevent loops, maintain connectivity, and recover from network failures. The goals of the lab today are the following:

- Build and verify a redundant Layer 2 network.
- Explain the effects of Ethernet loops and broadcast storms.
- Configure and validate the Spanning Tree Protocol (STP).
- Identify the root bridge and the roles of STP ports.
- Influence spanning-tree operation using bridge priority and path cost.
- Analyze STP behavior and convergence during topology changes and failures.

> **When your group finishes an exercise, make sure to alert the lecturer before continuing to the next exercise.**

[1] Read this lesson up to the section "*Putting it all together*." That final section is only a summary of the lesson, so you do **not** need to sign up to unlock it.

3 Exercise 1: Spanning tree in an access network

Build the Layer 2 network as shown Figure 1. The network consists of 6 switches. We will refer to the upper switches as distribution switches and the lower ones as access switches. Configure all IP addresses. All subnet masks are `/16` to guarantee full IP connectivity.

Follow the color coding shown in the figure for the cabling; this will make the next exercises easier (for example, use red for connections between hosts and access switches, and between access switches themselves, and blue for connections between access and distribution switches).

> **Leave one cable (e.g., the link between the access switches) disconnected to prevent creating a loop that could overload the network. After configuring Spanning Tree Protocol (STP), connect the final cable to complete the topology.**

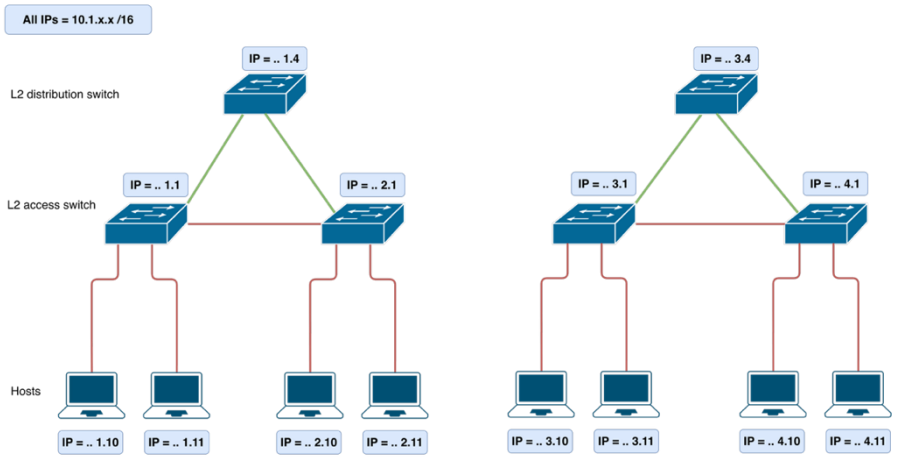


Figure 1: Spanning tree in the access network

> **IMPORTANT: Make sure you are disconnected from the campus network before starting the exercises.**

Now enable the Spanning Tree Protocol on all the devices and connect the last cable to close the loop. Don't forget to disable the firewall. Try to ping all the hosts and switches in the network using the `fping` command as demonstrated in lab session 2.

> **Tip: start a log visualizer before you make a loop in the network to analyze the traffic.**

Who can you reach?

_____

_____

Check the STP state. Can you identify the different states of the switches?

_____

_____

Who is the root, and why?

_____

What command can you use to display detailed STP info for the bridge port (switch)?

Use the command below to inspect the STP information of your switch:

```
brctl showstp <bridge name>
```

Study the bridge information and answer the following questions. What is the bridge ID of your switch? Write it down below.

A bridge ID is made up of multiple parts. Which part corresponds to the priority, and which part corresponds to the MAC address?

Why does the switch with the lowest bridge ID win the root bridge election?

Compare the bridge ID of your own switch with the designated root shown in the STP output. Is your switch the root bridge? Explain why or why not.

Which ports are blocking, and which ports are forwarding? Draw this on the schematic.

Open Wireshark and track the packets. Capture a BPDU and analyze its fields. Note down the Hello interval and the maximum age of the packet:

Do these values match what you saw in the values configured in the GUI?

Find out the real values by checking with a command through SSH. What command did you use? What did you find out?

Who is the root according to the BPDU packets?

_____

_____

Is a topology change indicated?

_____

_____

Now make the distribution switch the root (communicate with your peers about this event). How will you enforce this? Also, capture the packets using Wireshark!

_____

_____

Which ports are blocking, and which ports are forwarding? Draw this on the schematic.

_____

_____

4 Exercise 2: Spanning tree under link and switch failures

4.1 Distribution link down

Before starting this exercise, make sure your distribution switch is the root. Then **disconnect** an active distribution link and start a stopwatch.

How long does it take to complete the STP reconfiguration?

_____

_____

Check the STP protocol using the command `brctl showstp switch`. Who is the root, and what ports are blocked?

_____

_____

Repair the distribution link and start a stopwatch.

How long does it take to complete the STP reconfiguration

_____

_____

Check the STP protocol using the command `brctl showstp switch`. Who is the root, and what ports are blocked?

_____

_____

4.2 Access link down

Repeat the previous exercise, but instead **disconnect** an active link from an access switch.

Check the timing and the STP protocol.

_____

_____

Repair the link and check again.

_____

_____

4.3 Access switch down

Repeat the previous exercise, but instead **shut down** an access switch.

What is the difference with previous exercises?

_____

_____

4.4 No spanning tree

Remove the loop in the network and disable the spanning tree on all switches. Now put the loop back in there.

What happens when you try to ping?

_____

_____

5 Exercise 3: Spanning tree in a core network

Extend the Layer 2 network of the previous setup, as shown in Figure 2. Make it **fully resilient** by connecting the core switches in a resilient 'square'. Configure all IP addresses and use subnet mask `/16`. Make sure there is full connectivity and that STP is enabled.

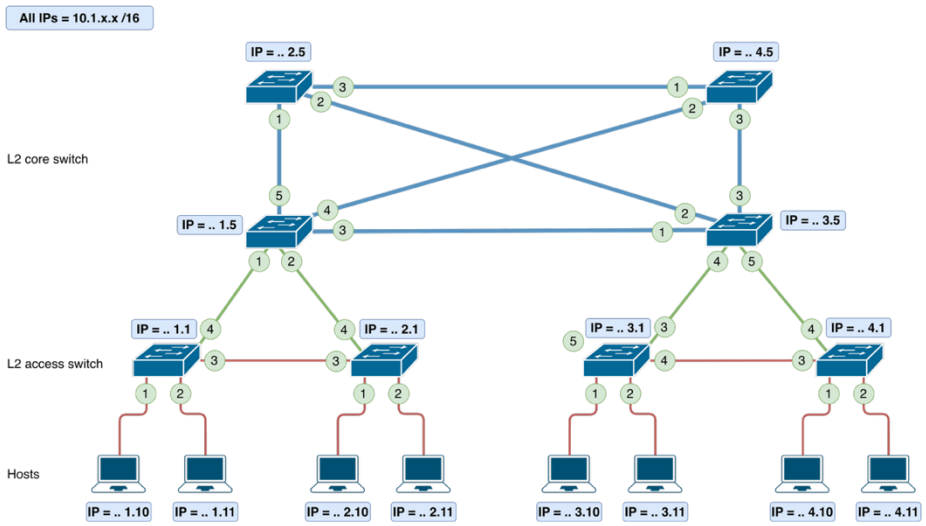


Figure 2: Spanning tree in the core network

After you have achieved full connectivity, check the STP.

> **Tip: Don't ping during the convergence of this bigger scheme. The switches have limited processing resources, and therefore it's best not to send too much additional data over the network during the STP convergence.**

Check the STP protocol using the command `brctl showstp switch`. Who is the root, which ports are forwarding/blocking/root ports/designated ports, and why?

_____

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Draw your spanning tree on the schematic.

_____

_____

Make one of the core switches the root (communicate with your peers about this event). Check the reconfiguration.

_____

_____

Are there any paths of equal cost? If yes, why and how was a certain path chosen?

_____

_____

5.1 Root switch down

Imagine the root switch fails. Who becomes the new root, and how is that determined?

_____

_____

Make sure that before you shut down the root switch, set up the configurations such that another core switch will become the new root. How will you do this?

_____

_____

Now shut down the root switch. How long does it take until the network converges again?

_____

_____

Ping all hosts and switches. Do you get all replies?

_____

_____

Check the STP. Did the secondary root switch become the new root?

_____

_____

Which ports are forwarding/blocking/root ports/designated ports, and why? Draw this on the new schematic.

_____

_____

Reboot the original root switch. What happens? Who becomes the new root?

_____

_____

Extra exercise: when you finish the lab, you can try to add even more loops to see when the switches give out. You can use Figure 3 as an example configuration. Why do you think the bridges cannot handle too many loops?

_____

_____

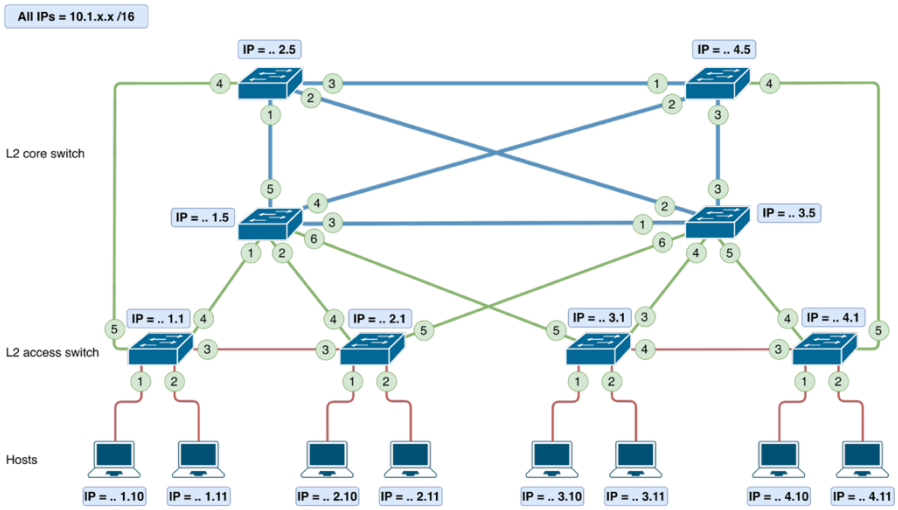


Figure 3: Possible configuration with extra loops

References

[1] NetworkAcademy.io, "Why do we need Spanning-Tree?," [Online]. Available: https://www.networkacademy.io/ccna/spanning-tree/why-do-we-need-stp.

[2] NetworkAcademy.io, "The Spanning-Tree Algorithm," [Online]. Available: https://www.networkacademy.io/ccna/spanning-tree/the-spanning-tree-algorithm.

[3] NetworkAcademy.io, "Spanning-Tree Port Roles," [Online]. Available: https://www.networkacademy.io/ccna/spanning-tree/spanning-tree-port-roles .

[4] NetworkAcademy.io, "STP Port States and Timers," [Online]. Available: https://www.networkacademy.io/ccna/spanning-tree/spanning-tree-states-and-timers.

[5] NetworkAcademy.io, "STP Topology Changes (TCN)," [Online]. Available: https://www.networkacademy.io/ccna/spanning-tree/stp-topology-changes.

[6] GeeksforGeeks, "Types of Spanning Tree Protocol (STP)," 9 December 2025. [Online]. Available: https://www.geeksforgeeks.org/computer-networks/types-of-spanning-tree-protocol-stp/.

[7] PowerCert Animated Videos, "Spanning Tree Protocol | CCNA - Explained," 4 November 2024. [Online]. Available: https://www.youtube.com/watch?v=6MW5P6Ci7lw.